

BLUESTOCK FINTECH
MUTUAL FUND ANALYTICS PLATFORM

End-to-End Data Engineering, ETL Pipeline &
Interactive Power BI Dashboard

CAPSTONE PROJECT REPORT



Submitted by

PRABHASH KUMAR

Data Analyst Intern

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Technologies

Python | SQL | SQLite | Power BI | Pandas | Git

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PRABHASH KUMAR

ABSTRACT

The Mutual Fund Analytics project was developed as part of the Data Analyst Internship at Bluestock Fintech Pvt. Ltd. The primary objective of this project is to build an end-to-end data analytics solution that collects, processes, analyzes, and visualizes mutual fund data to generate meaningful business insights. The project follows a complete ETL (Extract, Transform, Load) pipeline where multiple CSV datasets containing mutual fund information such as NAV history, AUM, SIP inflows, scheme performance, investor transactions, and portfolio holdings are processed using Python. Data cleaning, validation, transformation, and normalization techniques are applied to improve data quality before storing the processed data in an SQLite relational database. SQL queries are used to retrieve and analyze the stored data, while Power BI dashboards provide interactive visualizations of fund performance, investor behavior, market trends, and key financial indicators. Advanced financial analytics including CAGR, Sharpe Ratio, Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratio, Investor Cohort Analysis, SIP Continuity Analysis, and Fund Recommendation have been implemented to support informed investment decisions. The project demonstrates practical applications of Python, SQL, SQLite, and Power BI in financial data analytics and Business Intelligence. It provides an efficient, scalable, and interactive analytical platform that assists users in evaluating mutual fund performance and making data-driven investment decisions.

Keywords: Mutual Fund Analytics, ETL Pipeline, Python, SQL, SQLite, Power BI, Financial Analytics, Business Intelligence.

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